Simulation Analytics — Metric definitions

Simulated period 2025-01-02 02:39 → 2025-01-31 21:14

Metric	Definition	Formula / source
Incoming Trains KPIs		
Entry Queue wait	Time the train sat in the entry queue waiting for an arrival track to free up. Pure queue time — does NOT include the drive.	Incoming Trains!M = (E − D) = ArrivalDriveStart − EntryToStation
Arrival drive duration	Time the train spent physically driving from the entry gate to its assigned arrival track.	Incoming Trains!N = (F − E) = AtArrivalTrack − ArrivalDriveStart
ITP wait	Time between ITP being submitted to ResourceCU and the work actually commencing. Captures worker + shunt-loco allocation + travel-to-site time.	Incoming Trains!O = (H − G) = ITP_Starts − ITP_Init
ITP duration	Time the ITP work itself ran, once resources arrived.	Incoming Trains!P = (I − H) = ITP_Ends − ITP_Starts
PushOff wait	Time between PushOff being submitted and the work actually commencing. Captures worker re-allocation + passage-track wait.	Incoming Trains!Q = (K − J) = PushOff_Starts − PushOff_Init
PushOff duration	Time the entire PushOff envelope ran (all sub-drives + loco pull-backs between them).	Incoming Trains!R = (L − K) = PushOff_Ends − PushOff_Starts
Total in arrival yard	Total time the inbound train physically occupied an arrival track (from arrival to last WG pushed off).	Incoming Trains!S = (L − F) = PushOff_Ends − AtArrivalTrack
Total in system	End-to-end inbound time: from entry-queue arrival until PushOff completed.	Incoming Trains!T = (L − D) = PushOff_Ends − EntryToStation
Wagon Groups KPIs		
Time in arrival yard	Time the WG was on the arrival track as part of its parent train, from arrival to being pushed off.	Wagon Groups!M = (G − E) = PushOffDriveEnds − AtArrivalTrack
Time in classification yard	Time the WG sat on its classification track, from arriving there (= PushOffDrive ends) until the parent OBT physically departed.	Wagon Groups!N = (L − G) = DepartureDriveEnds − PushOffDriveEnds
Idle time on classification track	Time the WG waits after getting coupled or secured until the OBTP activity starts — the 'doing nothing' window after its own prep is done but before OBT-wide work begins.	Wagon Groups!P = (I − O) = OBTP_Starts − SEC/COP_End
Classification context	Plain-language narrative of the WG's time on the classification track: how it joined (secured first, or coupled to the sibling already there), the wait before each later sibling coupled, the OBTP window, and departure.	Wagon Groups!Q; built in parse_wagon_groups.
Outbound Trains KPIs		
OBTP wait	Time between OBTP being submitted and actually starting. Captures worker + train-loco allocation + travel.	Outbound Trains!K = (F − E) = OBTP_Starts − OBTP_Init
OBTP duration	Time the OBTP work itself ran.	Outbound Trains!L = (G − F) = OBTP_Ends − OBTP_Starts
DEPD gate wait	Time the OBT was ready to depart (DEPD submitted) but stuck waiting for the single exit-gate resource. Headline bottleneck metric given exitGateCount = 1.	Outbound Trains!M = (I − H) = DEPD_Starts − DEPD_Init
DEPD duration	Time the train spent physically driving from its classification track to the exit gate.	Outbound Trains!N = (J − I) = DEPD_Ends − DEPD_Starts
Total on classification track	End-to-end outbound time, from OBT-creation moment (≈ OBTP init) until DEPD completes.	Outbound Trains!O = (J − E) = DEPD_Ends − OBTP_Init
Activity type glossary		
Activity type abbreviations	ARRD = ArrivalDrive (train enters yard on own loco); ITP = IncomingTrainPreparation (workers prep arriving train); PO = PushOff (parent envelope); POD = PushOffDrive (one sub-drive per consecutive-same-destination cut of WGs); SEC = Securing (first WG on a classification track); COP = Coupling (joining WG to existing WGs on track); OBTP = OutboundTrainPreparation (workers + train loco prep OBT); DEPD = DepartureDrive (OBT exits through gate).	(reference — see Models/Activity.cs GetActivityAbbreviation).
WG → OBT matching		
How parent OBT is derived for a WG	WGs are matched to OBTs by classification track + time window: an OBT created at time T on track X owns every WG that arrived on track X at or before T and wasn't already claimed by an earlier OBT on the same track. Greedy, earliest-OBT-first.	parse_obts_and_wgs in analytics.py.

Summary tables

Durations in h:mm:ss. N = instances with both timestamps present.

Throughput

Metric	Value	Per day
Simulated period	2025-01-02 02:39 → 2025-01-31 21:14	
Duration	29.8 days	
Inbound trains entered	331	11.1
Inbound trains pushed off	331	11.1
Wagon groups (total)	1189	39.9
Wagon groups departed	1038	34.9
Outbound trains created	426	14.3
Outbound trains departed	410	13.8
Activities logged	3981	

Process durations

Activity type	N	Mean	Median	Min	Max	Std
ArrivalDrive	331	0:03:10	0:03:17	0:01:47	0:04:02	0:00:31
IncomingTrainPreparation	331	0:07:32	0:04:35	0:00:10	0:32:13	0:06:48
PushOff	331	0:08:31	0:04:01	0:00:00	0:37:23	0:08:40
PushOffDrive	1084	0:02:36	0:02:42	0:01:31	0:04:28	0:00:32
Securing	525	0:02:09	0:02:12	0:02:00	0:02:12	0:00:05
Coupling	559	0:04:17	0:04:24	0:04:00	0:04:24	0:00:11
OutboundTrainPreparation	410	0:51:32	0:41:27	0:25:07	2:42:57	0:26:27
DepartureDrive	410	0:01:24	0:01:16	0:00:53	0:02:52	0:00:22

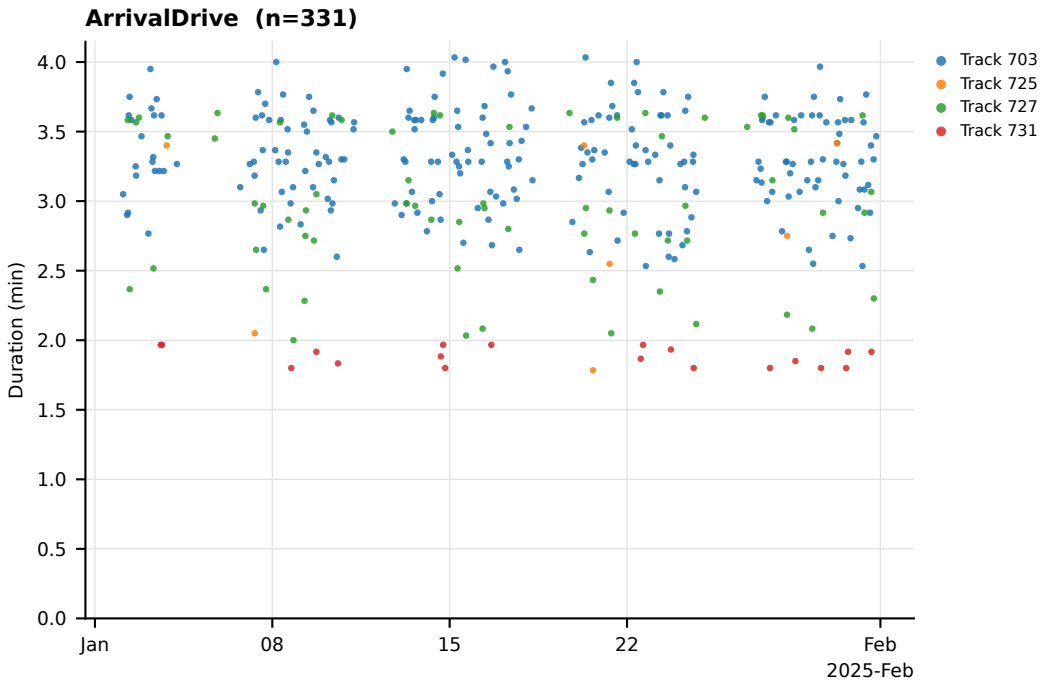
Train timeline (inbound trains)

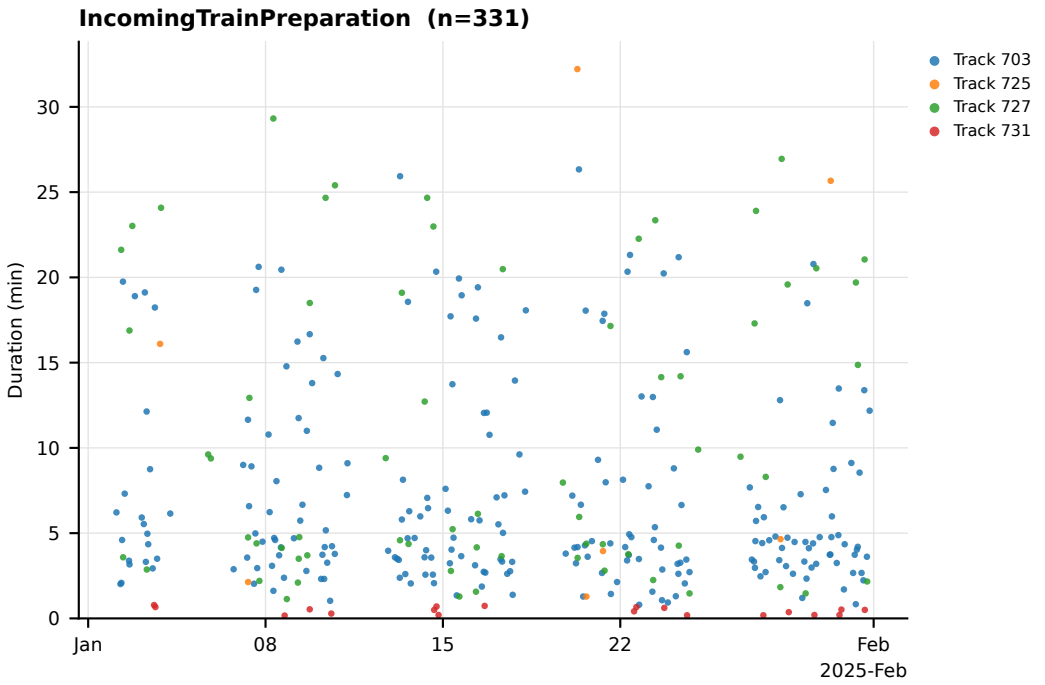
Metric	N	Mean	Median	Min	Max
Entry queue wait	331	0:00:00	0:00:00	0:00:00	0:00:00
Arrival drive	331	0:03:10	0:03:17	0:01:47	0:04:02
ITP wait	331	0:04:00	0:04:00	0:04:00	0:04:00
ITP duration	331	0:07:32	0:04:35	0:00:10	0:32:13
PushOff wait	331	0:02:27	0:01:15	0:01:03	0:26:56
PushOff duration	331	0:08:31	0:04:01	0:00:00	0:37:23
Total in arrival yard	331	0:22:30	0:14:44	0:05:20	1:11:57
Total in system	331	0:25:40	0:18:05	0:07:08	1:15:31

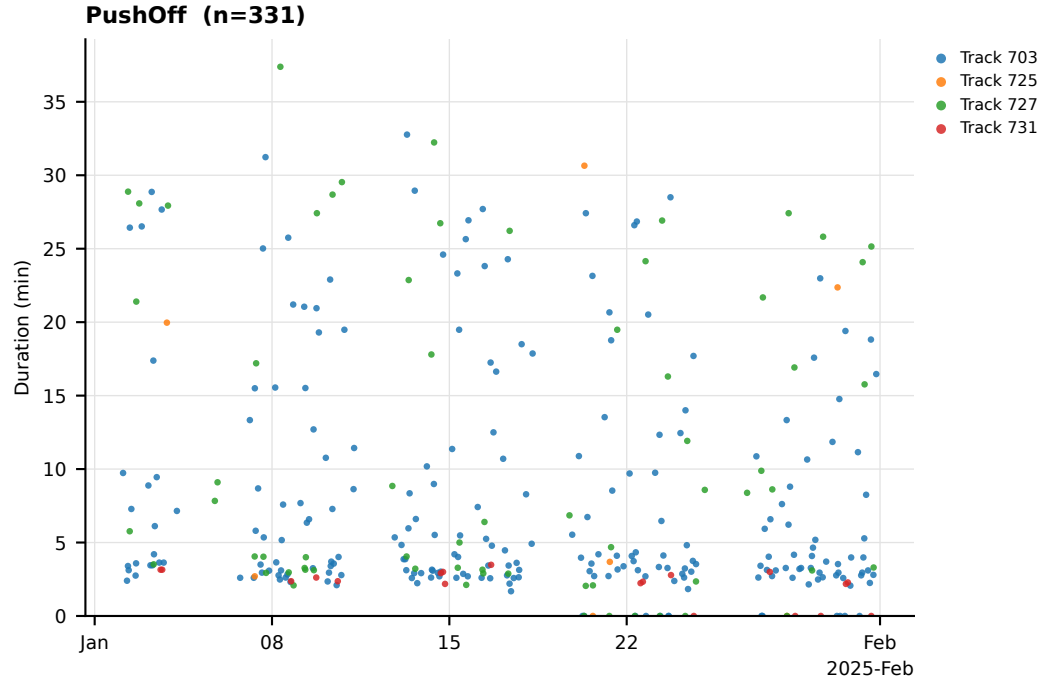
OBT / WG formation

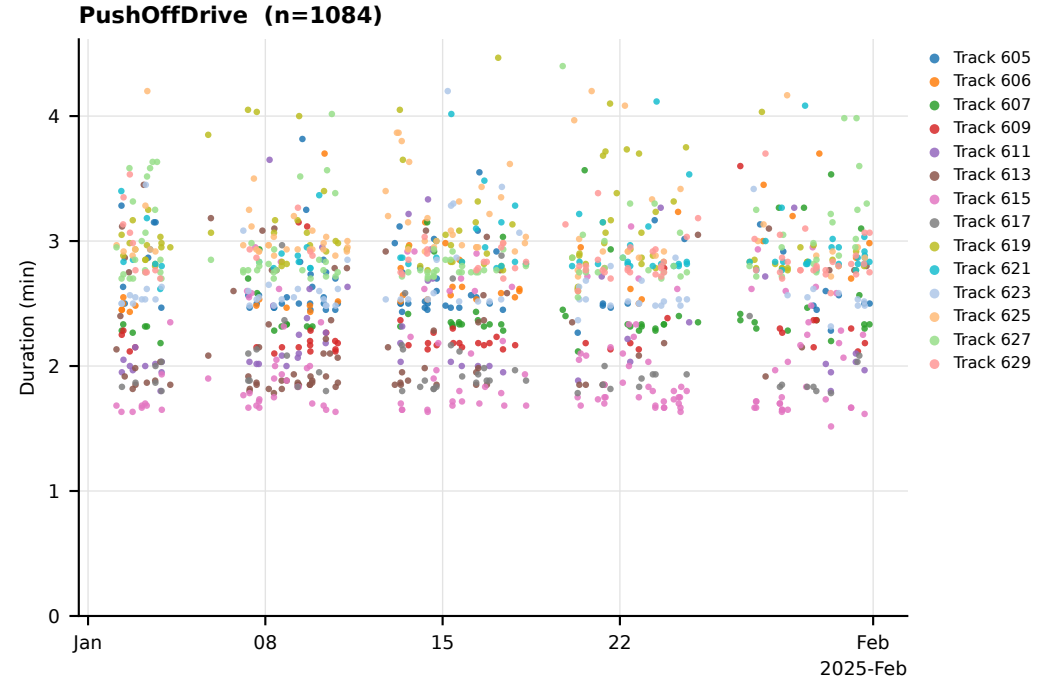
Metric	N	Mean	Median	Min	Max
Outbound trains (per OBT)					
OBTP wait	410	11:00:43	0:25:00	0:01:08	72:31:19
OBTP duration	410	0:51:32	0:41:27	0:25:07	2:42:57
DEPD gate wait	410	0:00:01	0:00:00	0:00:00	0:01:18
DEPD duration	410	0:01:24	0:01:16	0:00:53	0:02:52
Total on classification track	410	11:53:40	1:39:11	0:27:30	73:39:15
Wagon groups (per WG)					
Time in arrival yard	1088	0:30:43	0:30:33	0:07:38	1:11:57
Time in classification yard	1038	22:52:55	16:42:06	0:30:54	166:42:07
Idle on classification track	1038	21:59:09	15:49:32	0:01:08	166:09:21

Activity durations by track — duration (min) vs start time (1/2)

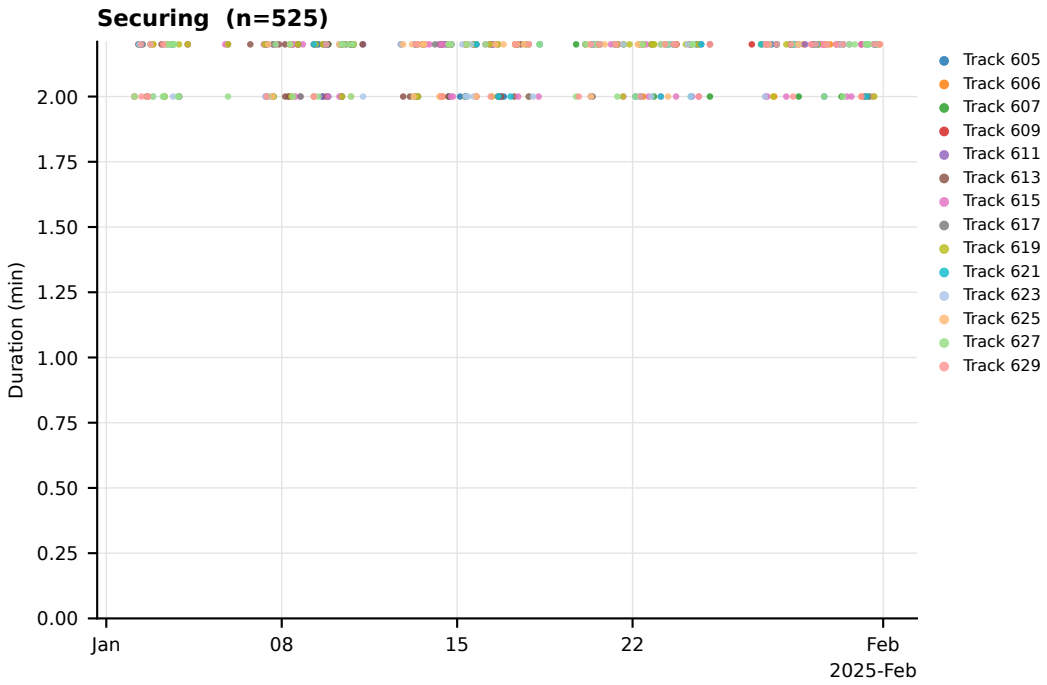


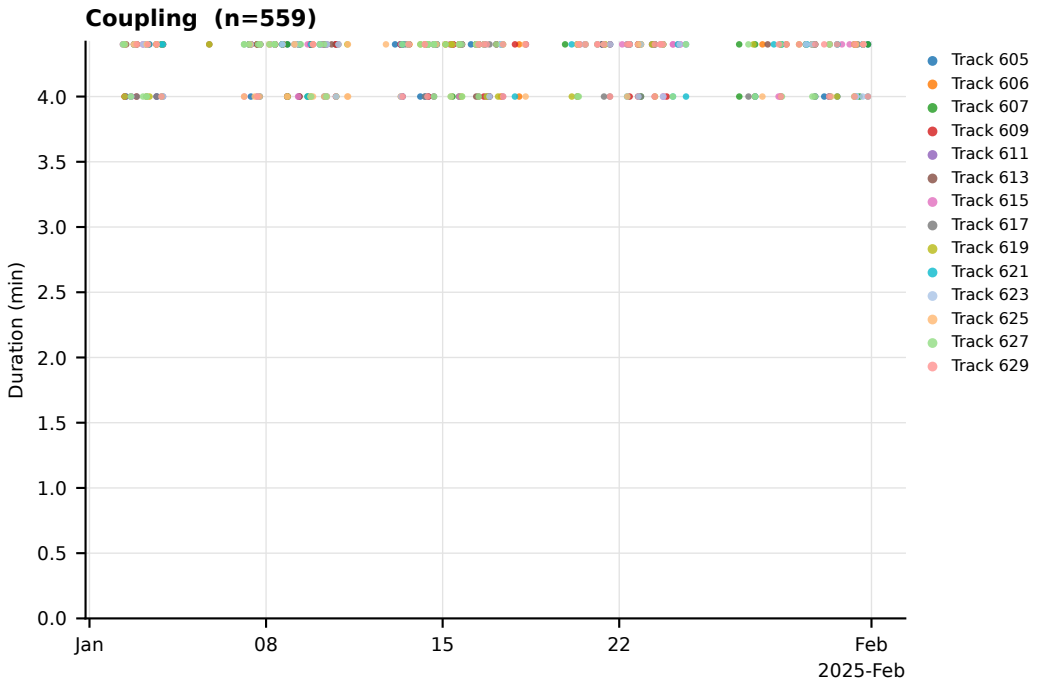


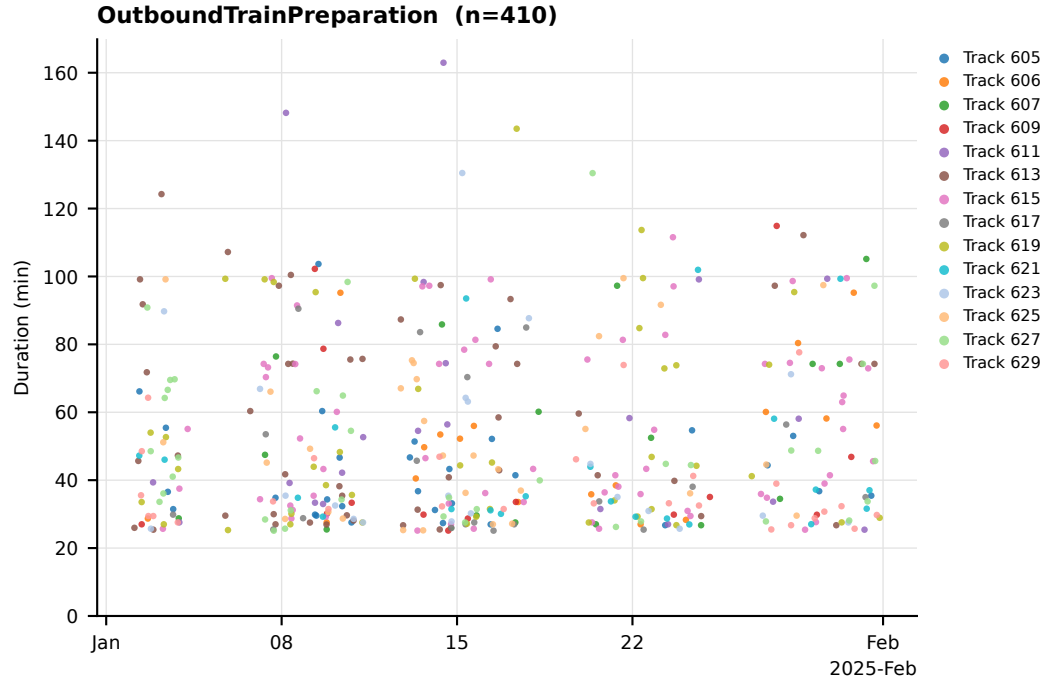


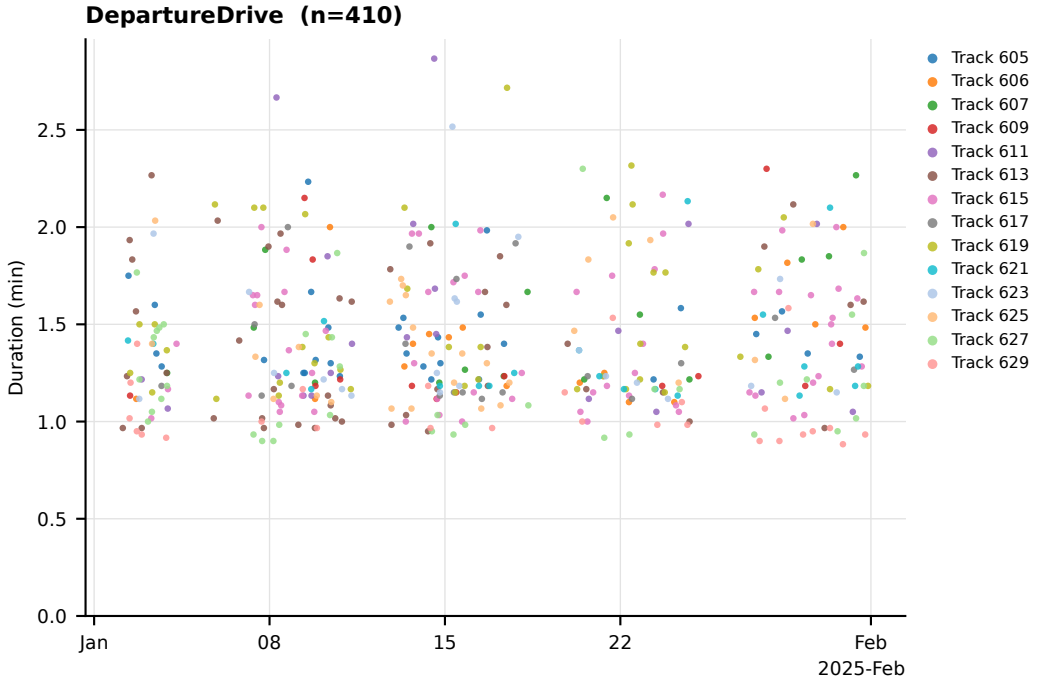


Activity durations by track — duration (min) vs start time (2/2)

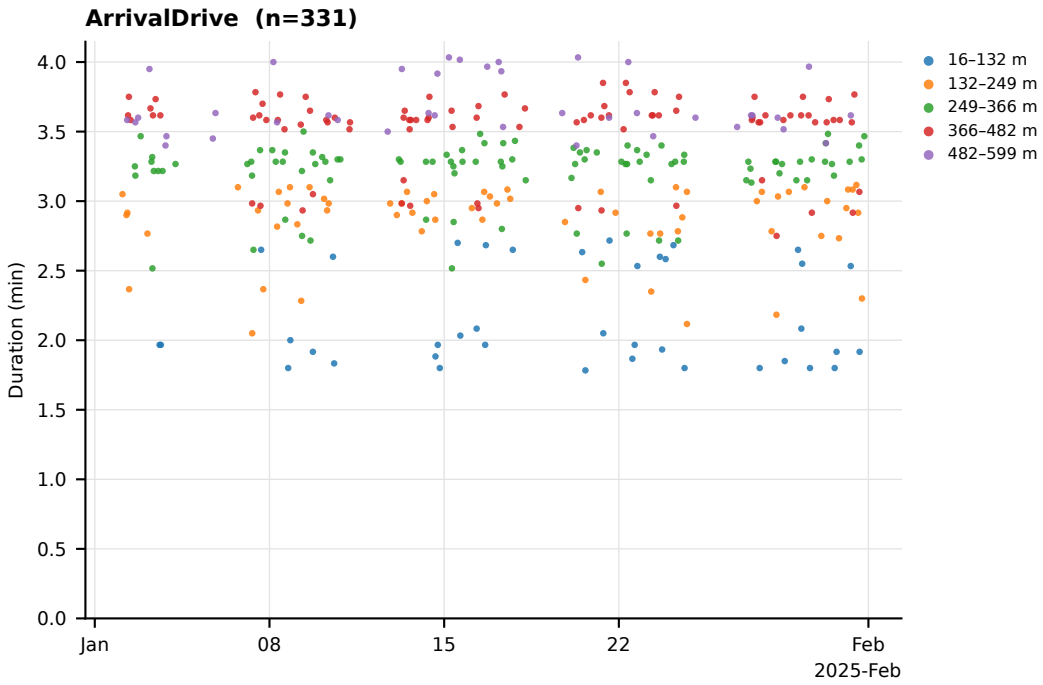


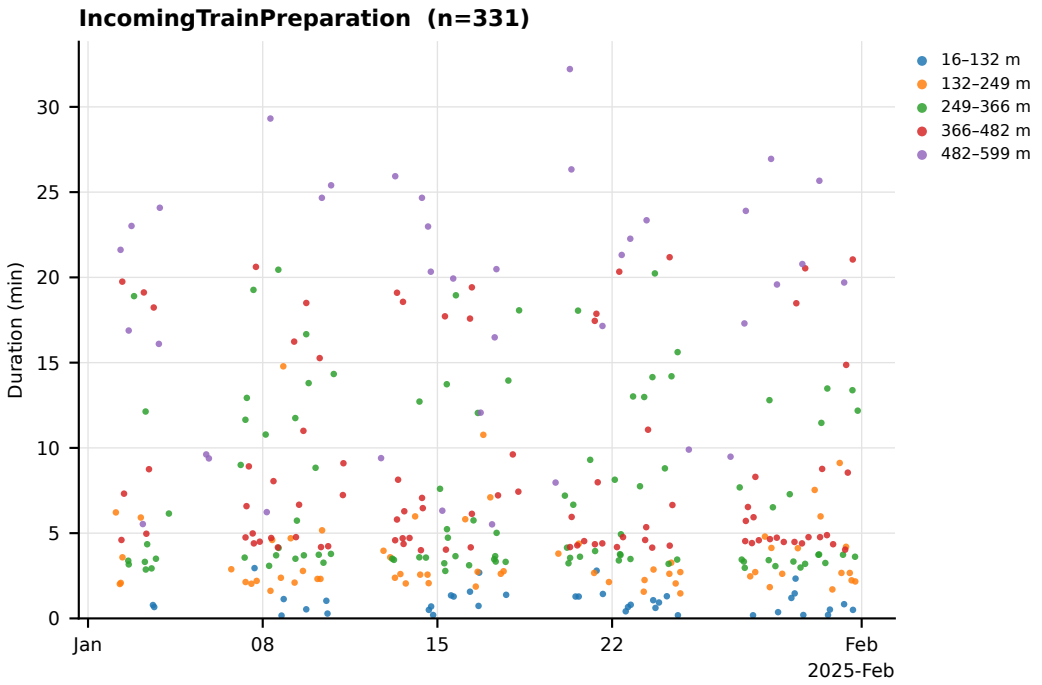


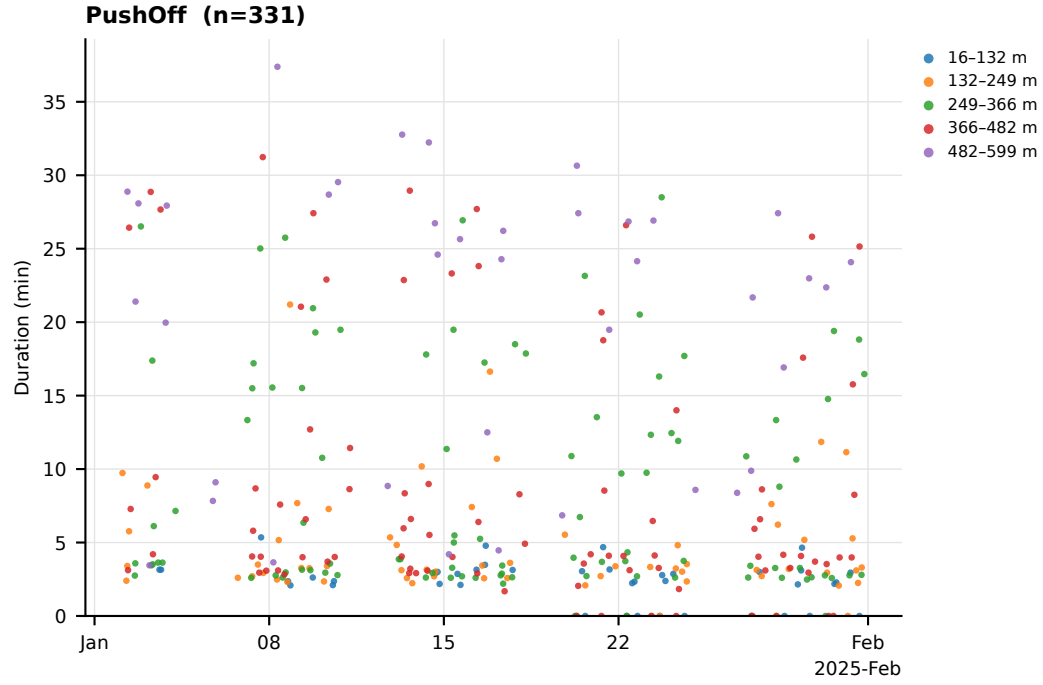


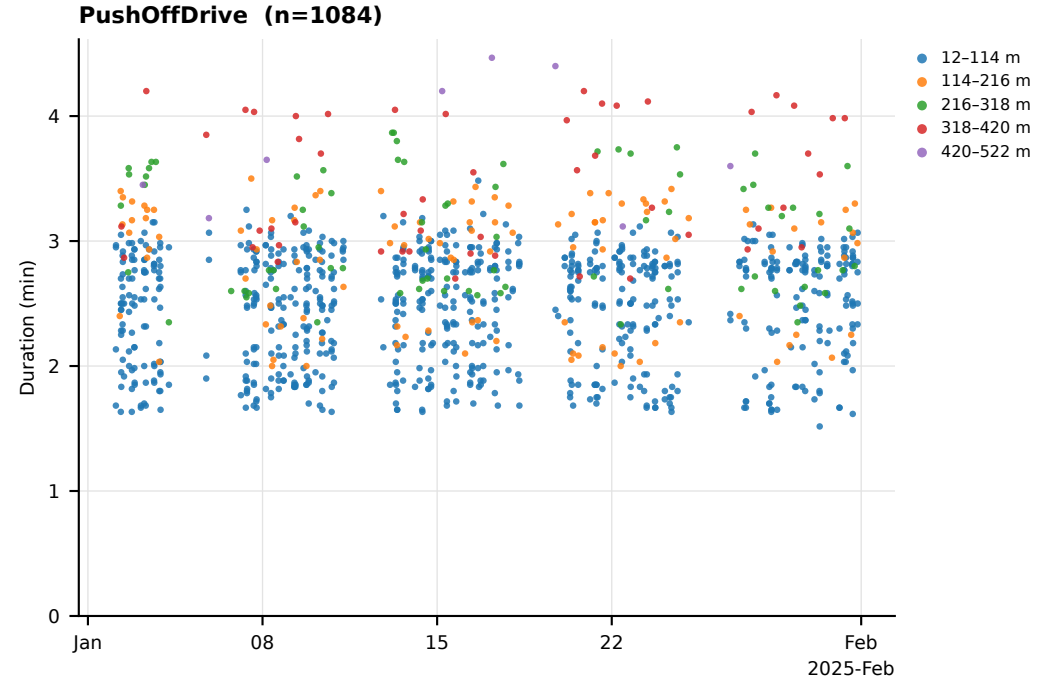


Activity durations by length — duration (min) vs start time (1/2)









Activity durations by length — duration (min) vs start time (2/2)

